

Course Code- 151602 Automata Theory & Compiler Design**3 0 0 3****Unit-1.0****7 hrs.**

Formal Language and Regular Expressions : Languages, Definition Languages regular expressions, Finite Automata – DFA, NFA. Conversion of regular expression to NFA, NFA to DFA.

Unit-2.0**7 hrs.**

Applications of Finite Automata to lexical analysis, lex tools. Context Free grammars and parsing : Context free grammars, derivation, parse trees, ambiguity LL(K) grammars and LL(1) parsing.

Unit-3.0**7 hrs.**

Bottom up parsing handle pruning LR Grammar Parsing, LALR parsing, parsing ambiguous grammars, YACC programming specification. Semantics : Syntax directed translation, S-attributed and L-attributed grammars, Intermediate code – abstract syntax tree, translation of simple statements and control flow statements.

Unit-4.0**7 hrs.**

Context Sensitive features – Chomsky hierarchy of languages and recognizers. Type checking, type conversions, equivalence of type expressions, overloading of functions and operations.

Unit-5.0**7 hrs.**

Run time storage : Storage organization, storage allocation strategies scope access to now local names, parameters, language facilities for dynamics storage allocation. Code optimization: Principal sources of optimization, optimization of basic blocks, peephole optimization, flow graphs, Data flow analysis of flow graphs.

Unit-6.0**7 hrs.**

Code generation : Machine dependent code generation, object code forms, generic code generation algorithm, Register allocation and assignment. Using DAG representation of Blocks.

Text/Reference:

1. Introduction to Theory of computation.Sipser, 2nd Edition, Thomson.
2. Compilers Principles, Techniques and Tools Aho, Ullman, Ravisethi, Pearson Education.
3. Modern Compiler Construction in C , Andrew W.Appel Cambridge University Press.
4. Compiler Construction, LOUDEN, Thomson.
5. Elements of Compiler Design, A. Meduna, Auerbach Publications, Taylor and Francis Group.
6. Principles of Compiler Design, V. Raghavan, TMH.
7. Engineering a Compiler, K. D. Cooper, L. Torczon, ELSEVIER.